DAY 06 | PRE-READ

Forms, Rules and Translators

A light read before Day 6. About 15 minutes, no code required.

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Why read this?

On Day 5 you built your first API. Tomorrow you will make it safer and more organised, and connect it to a real database. Three everyday ideas explain almost everything you will see.

1. Where information comes from

Imagine you call a hotel. You might say a room number ("room 204"), add preferences ("a quiet room, high floor"), or hand over a filled-in form (your name, dates, number of guests). APIs receive information in the same three ways:

Hotel example	API name	Used for
Room 204	Path parameter	Pointing to one specific thing
Quiet, high floor	Query parameter	Filtering and preferences
A filled-in booking form	Body	Sending a full set of details

2. A form with rules

A good booking form refuses a check-out date before the check-in date, and will not accept "abc" as the number of guests. These rules protect the hotel from mistakes. In FastAPI, a tool called **Pydantic** lets you write these rules once. Bad data is turned away at the door, with a clear message explaining why.

> **Key idea:** describe the shape of your data once, and let the tool check every request for you.

3. Organising a growing menu

A small cafe has one menu page. A big restaurant splits it into sections: starters, mains, desserts. APIs do the same with **routers**: one section for books, one for members, one for loans.

4. A translator between two languages

Python thinks in objects. Most databases think in tables. An **ORM** (Object Relational Mapper) is a translator between them. You work with Python objects, and the ORM writes the database commands for you. We use one called SQLAlchemy.

Python side	Database side
Class	Table
Attribute	Column
Object	Row

5. Think about these before class

- Think of a form you filled in recently. Which fields were required, and which had rules?
- If you were designing a library app, what sections would its menu have?
- Why might a translator make mistakes less likely than speaking a language you barely know?

Kihon ga daiji (基本が大事): the basics are what matter most